



**|| Jigjiga University || Institute of Technology || Department of Information Technology||**

**|| Model Exam || Date: 06/06/2023 ||**

Time: 100 Minute (1hour: 40Minute)

**Name of the student:** \_\_\_\_\_ **ID. No:** \_\_\_\_\_  
**Dept. IT**

**Instruction I: Multiple Choice Questions (Tick the correct alternative)**

1. HTML is the standard \_\_\_\_ language for creating Web pages.
  - A. scripting
  - B. programming
  - C. styling
  - D. markup
2. Which tag is the root element of an HTML page?
  - A. <html>
  - B. <title>
  - C. <head>
  - D. <body>
3. Who invented HTML?
  - A. Dave Raggett
  - B. Tim Berners-Lee
  - C. Denis Ritchie
  - D. All of the above
4. HTML tags with no content are called \_\_\_\_\_.
  - A. Special tags
  - B. Advanced tags
  - C. Empty tags
  - D, Other tags
5. HTML stands for:

- A. HyperText Markup Language
  - B. Hyperlink Text Markup Language
  - C. High-level Text Markup Language
  - D. Home Tool Markup Language
6. Which of the following is an HTML tag for creating a heading?
- A. <title>
  - B. <header>
  - C. <h1>
  - D. <p>
7. Which HTML element is used to define a hyperlink?
- A. <link>
  - B. <a>
  - C. <href>
  - D. <hyperlink>
8. The <table> element is used to create:
- A. Ordered lists
  - B. Unordered lists
  - C. Tables
  - D. Divisions
9. The <form> element is used to:
- A. Define a section of the web page
  - B. Create a table
  - C. Display an image
  - D. Create an interactive form for user input
10. The <iframe> element is used to:
- A. Display an image
  - B. Create a hyperlink
  - C. Embed content from another source
  - D. Define a section of the web page
11. The <footer> element is typically used to:
- A. Define a section of the web page
  - B. Create a table

- C. Display an image
  - D. Provide information about the author or copyright of the web page
12. Which PHP function is used to establish a connection with a MySQL database?
- A. `mysql_connect()`
  - B. `mysqli_connect()`
  - C. `connect_mysql()`
  - D. `mysql_db_connect()`
13. What is the correct way to include an external PHP file named "functions.php"?
- A. `require("functions.php");`
  - B. `include("functions.php");`
  - C. `require_once("functions.php");`
  - D. All of the above
14. How can you check the length of a string variable in PHP?
- A. `length($str)`
  - B. `strlen($str)`
  - C. `strlength($str)`
  - D. `size($str)`
15. Which PHP function is used to redirect the user to a different URL?
- A. `redirect()`
  - B. `header()`
  - C. `forward()`
  - D. `navigate()`
16. What is the correct way to start a PHP session?
- A. `session_start();`
  - B. `start_session();`
  - C. `initiate_session();`
  - D. `begin_session();`

17. Which of the following is used to write a message to the server error log in PHP?

- A. `log_message()`
- B. `error_log()`
- C. `write_log()`
- D. `server_log()`

18. Which of the following is the correct way to declare a constant in PHP?

- A. `define("CONSTANT_NAME", "value");`
- B. `constant("CONSTANT_NAME", "value");`
- C. `const CONSTANT_NAME = "value";`
- D. `var CONSTANT_NAME = "value";`

19. How do you get the current date and time in PHP?

- A. `get_date()`
- B. `current_datetime()`
- C. `date_time()`
- D. `date()`

20. What is the purpose of the PHP function "isset()".

- A. To check if a variable is set and not NULL
- B. To check if a variable has a specific value
- C. To check if a variable is empty
- D. To check if a variable is numeric

21. What is the correct way to handle file uploads in PHP?

- A. Use the `$_POST` array to retrieve the uploaded file data
- B. Use the `$_GET` array to retrieve the uploaded file data
- C. Use the `$_FILE` array to retrieve the uploaded file data
- D. Use the `$_REQUEST` array to retrieve the uploaded file data

22. Which of the following is NOT a valid PHP data type?

- A. `boolean`

- B. integer
- C. float
- D. string

23. How can you include a file in PHP without generating a warning if the file is not found?

- A. `include_once("file.php");`
- B. `include("file.php");`
- C. `require_once("file.php");`
- D. `require("file.php");`

24. Which of the following is the correct way to add a comment in PHP code?

- A. `#`
- B. `//`
- C. `/**/`
- D. All Mentioned

25. Which of the following is the default file extension of PHP files?

- A. `.Php`
- B. `.Ph`
- C. `.Xml`
- D. `.Html`

26. What will be the output of the following PHP code?

```
<?php
$x = 10;
$y = 20;
if ($x > $y && 1||1)
    print "JJU IT department" ;
else
    print "Welcome to Sanfoundry";
?>
```

- A. no output

- B. Welcome to Sanfoundry
- C. JJU IT department
- D. Error

27. Which is the right way of declaring a variable in PHP?

- A. \$3hello
- B. \$\_hello
- C. \$this
- D. \$5\_Hello

28. A function in PHP which starts with \_\_ (double underscore) is known as

- A. Default Function
- B. User Defined Function
- C. Inbuilt Function
- D. Magic Function

29. Which of the following web servers are required to run the PHP script?

- A. Apache and PHP
- B. IIS
- C. XAMPP
- D. Any of the mentioned

30. Which of the following variables does PHP use to authenticate a user?

- i) \$\_SERVER['PHP\_AUTH\_USER'].
- ii) \$\_SERVER['PHP\_AUTH\_USERS'].
- iii) \$\_SERVER['PHP\_AUTH\_PU'].
- iv) \$\_SERVER['PHP\_AUTH\_PW'].

- A. ii) and iv)
- B. i) and iv)
- C. ii) and iii)
- D. i) and ii)

31. Which programming language is commonly used for developing Android applications?

- A. Java
- B, C#

- C. Swift
  - D. Python
32. What is the main purpose of an Activity in Android development?
- A. Managing background tasks
  - B. Handling user interactions
  - C. Storing data locally
  - D. Performing network operations
33. Which of the following is an iOS-specific programming language?
- A. Java
  - B. C#
  - C. Swift
  - D. Kotlin
34. What does SDK stand for in mobile application development?
- A. Software Development Knowledge
  - B. Software Development Kit
  - C. System Development Kit
  - D. System Development Knowledge
35. What is the primary function of a layout in mobile app development?
- A. Handling user input
  - B. Managing data storage
  - C. Displaying the user interface
  - D. Performing network operations
36. Which of the following is a cross-platform mobile app development framework?
- A. Android Studio
  - B. Xcode
  - C. Flutter
  - D. React Native
37. What is the purpose of Intent in Android development?
- A. Storing data locally
  - B. Handling user interactions
  - C. Launching activities and services

- D. Performing network operations
38. Which mobile platform uses Objective-C as its primary programming language?
- A. Android
  - B. iOS
  - C. Windows Phone
  - D. BlackBerry
39. Which tool is commonly used for iOS app development?
- A. Android Studio
  - B. Visual Studio
  - C. Xcode
  - D. Eclipse
40. What does API stand for in the context of mobile app development?
- A. Application Programming Interface
  - B. Android Programming Interface
  - C. Apple Programming Interface
  - D. Application Platform Interface
41. In case of any shut down during transaction before commit which of the following statement is done automatically?
- A. View
  - B. Commit
  - C. Rollback
  - D. Flashback
42. For designing a normal RDBMS which of the following normal form is considered adequate?
- A. 4NF
  - B. 3NF
  - C. 2NF
  - D. 5NF
43. What is the function of the following command?
- Delete from Employee where Salary =5000;*
- A. Clears entries from relation
  - B. Deletes relation
  - C. Deletes particular tuple from relation
  - D. All of the mentioned

44. Which command is used to remove a relation from an SQL?
- A. Drop table
  - B. Delete
  - C. Purge
  - D. Remove
45. The ability to query data, as well as insert, delete, and alter tuples, is offered by
- A. TCL (Transaction Control Language)
  - B. DCL (Data Control Language)
  - C. DDL (Data Definition Language)
  - D. DML (Data Manipulation Language)
46. Which of the following is not a type of database?
- A. Hierarchical
  - B. Network
  - C. Distributed
  - D. Decentralized
47. At \_\_\_\_ level of database design, all the database entities and the relationships among them are included.
- A. Physical
  - B. Conceptual
  - C. External
  - D. Internal
48. All are database security mechanisms, except,
- A. Encryption
  - B. Authentication
  - C. Authorization
  - D. SQL statement
49. Identify technique for implementing query optimization, which takes into account the contextual nature of user queries.
- A. Semantic-based Query Optimization
  - B. Heuristic-based Query Optimization
  - C. Cost-based Query Optimization
  - D. A and B
50. Among the following options identify the one which represents the correct steps of Query Processing.
- A. Input → output → processing → evaluation → execution → output
  - B. Output → parsing → translation → Optimizer → Evaluation → Execution
  - C. Input → parsing → translation → Optimizer → Evaluation → Execution
  - D. Optimizer → Input → parsing → translation → Optimizer → Evaluation → Execution
51. Point out the correct statement.
- A. It is not advisable to extract all data from large databases.

- B. It is advisable to extract only specific columns from large databases.
  - C. Using SELECT \* SQL code in large database does not requires more resources.
  - D. A and B
  - E. All
52. Among the following identify an encryption algorithm in which converting encrypted data to their original value is an impossible task.
- A. Hashing
  - B. Symmetric
  - C. Asymmetric
  - D. Digital Signatures

53. Which one of the options listed is not considered a goal of database security?
- A. Integrity
  - B. Availability
  - C. Confidentiality
  - D. Velocity
54. Which of the following is not a relational algebraic operation that is not form the set theory?
- A. UNION
  - B. INTERSECTION
  - C. CARTESIAN PRODUCT
  - D. SELECT
55. What will be the number of columns of CARTESIAN PRODCUT if the participating relations have 5 and 7 columns respectively?
- A. 35
  - B. 8
  - C. 7
  - D. 12
56. What is a database system?
- A. A collection of data that is organized and stored in a way that makes it easy to retrieve and use.
  - B. A software program that allows users to create, manage, and access data.
  - C. A physical device that stores data.
  - D. A set of rules that govern how data is stored and accessed.
  - E. A and B
57. Which of the following is true regarding centralized database and distributed database?
- A. A centralized database is stored on a single server, while a distributed database is stored on multiple servers.
  - B. A centralized database is more scalable than a distributed database.
  - C. A centralized database is more efficient than a distributed database.
  - D. A centralized database is more secure than a distributed database.
  - E. A and D
58. In the relational table, which of the following can also be represented by the term "attribute"?
- A. Entity
  - B. Row
  - C. Column
  - D. B & C
59. Which of the following commands is used to save any transaction permanently into the database?
- A. Commit
  - B. Rollback
  - C. Save point

- D. None of the above
60. What is the relation calculus?
- A. It is a kind of procedural language
  - B. It is a non-procedural language
  - C. It is a high-level language
  - D. It is Data Definition language
61. Which cryptographic algorithm forms the basis of the El Gamal cryptosystem?
- A. RSA
  - B. Diffie-Hellman
  - C. DES
  - D. IDEA
62. If Kedir wants to send an encrypted message to Halima using a public key cryptosystem, which key does he use to encrypt the message?
- A. Kedir's public key
  - B. Kedir's private key
  - C. Halima's public key
  - D. Halima's private key
63. What encryption technique does WPA2 use to protect wireless communications?
- A. TKIP
  - B. DES
  - C. 3DES
  - D. AES
64. Which standard defines the most commonly used format for public key certificates.
- A. X.500
  - B. X.509
  - C. X.900
  - D. X.905
65. Which of the following links would be protected by WPA (WiFi Protected Access) encryption?
- A. Firewall to firewall
  - B. Router to firewall
  - C. Client to wireless access point
  - D. Wireless access point to router
66. What does IPsec define?
- A. All possible security classifications for a specific configuration
  - B. A framework for setting up a secure communication channel
  - C. The valid transition states in the Biba model
  - D. TCSEC security categories

67. According to A general model of access control which of the following is true related to a subject?
- A. A subject is always a user account.
  - B. The subject is always the entity that provides or hosts the information or data.
  - C. The subject is always the entity that receives information about or data from an object.
  - D. A single entity can never change roles between subject and object.
68. Which of the following is not considered a violation of confidentiality?
- A. Stealing passwords
  - B. Eavesdropping
  - C. Hardware destruction
  - D. Social engineering
69. How can an individual/person best be identified or authenticated to prevent local masquerading attacks?
- A. User Id and password
  - B. Smart card and PIN code
  - C. Two-factor authentication
  - D. Biometrics
70. What is the length of a message digest produced by the MD5 algorithm?
- A. 64 bits
  - B. 128 bits
  - C. 256 bits
  - D. 384 bits
71. Which of the following would be a logical first step in troubleshooting a PC?
- A. Check the computer CMOS
  - B. Define the circumstances of the problem
  - C. Call the vendor
  - D. Define what applications are being used
72. Which would you do first when troubleshooting a faulty monitor?
- A. Check its connections to the computer and power source
  - B. Use a meter to check the CRT and internal circuitry for continuity
  - C. Power down the monitor, then turn it on again to see if that corrects the problem
  - D. Power down the computer, then turn it on again to see if that corrects the problem
73. You have a PC with no video\* which of the following is LEAST likely to be Causing the problem?
- A. Defective RAM (bank zero)
  - B. Defective microprocessor
  - C. Crashed hard drive
  - D. Loose video card
74. When connecting a ribbon cable to a connector, how do you know which Direction to plug it in

- A. The red line in the cable goes to the highest pin number
  - B. The colored line in the cable goes to pin #1
  - C. It does not matter
  - D. The blue or red line in the cable goes away from the power connector
75. What is the first step in diagnosing a completely dead computer at the client site that was working the day before?
- A. Test the power supply
  - B. Replace the CMOS battery
  - C. Check the AC outlet
  - D. reseal the hard drive controller cable
76. What is the difference between servlets and applets?
- i. Servlets execute on Server; Applets execute on browser
  - ii. Servlets have no GUI; Applet has GUI
  - iii. Servlets creates static web pages; Applets creates dynamic web pages
  - iv. Servlets can handle only a single request; Applet can handle multiple requests
- A. i, ii, iii are correct
  - B. i, ii are correct
  - C. i, iii are correct
  - D. i, ii, iii, iv are correct
77. How do you insert COMMENTS in Java code?
- A. // This is a comment
  - B. # This is a comment
  - C. /\* This is a comment
  - D. All
78. Which keyword is used to create a class in Java?
- A. class
  - B. className
  - C. class()
  - D. MyClass
79. What is the output of the following piece of code?
- ```

Public class array
{
    Public static void main(String args[])
    {
        int []arr = {1,2,3,4,5};
        System.out.println(arr[2]);
        System.out.println(arr[4]);
    }
}

```
- A. 3 and 5
  - B. 5 and 3
  - C. 2 and 4

D. 4 and 2

80. What will be the output of below statements?

```
String s = "Java Programming"+9+3+"Exam"+(8+2);
```

```
System.out.println(s);
```

- A. Java Programming12Exam10
- B. Java Programming93Exam10
- C. Java Programming93Exam82
- D. Java Programming12Exam82

81. Which statement is true about Java?

- A. Java is a sequence-dependent programming language
- B. Java is a code dependent programming language
- C. Java is a platform-dependent programming language
- D. Java is a platform-independent programming language

82. Which component is used to compile, debug and execute the java programs?

- A. JRE
- B. JIT
- C. JDK
- D. JVM

83. Which one of the following is not a Java feature?

- A. Object-oriented
- B. Use of pointers
- C. Portable
- D. Dynamic and Extensible

84. Which of the following statements correctly differentiates the structural and object-oriented programming paradigms?

- A. In structural programming, programs are organized around objects and their interactions, while object-oriented programming focuses on functions and procedures.
- B. Both structural and object-oriented programming paradigms organize programs around objects and their interactions.
- C. In structural programming, programs are organized around functions, while object-oriented programming organizes programs around objects and their interactions.
- D. Both structural and object-oriented programming paradigms organize programs around functions and procedures.

85. Which exception is thrown when an array is accessed with an invalid index?
- A. `ArrayIndexOutOfBoundsException`
  - B. `NullPointerException`
  - C. `ArithmeticException`
  - D. `IllegalArgumentException`
86. From the following protocols the one is properly matched with its port number.
- A. FTP port number is 53
  - B. HTTP port number is 443
  - C. HTTPS port number is 80
  - D. DNS port number is 53
87. If \_\_\_\_\_ command is issued; files are transferred from the server to the clients.
- A. Get
  - B. Put
  - C. Both
  - D. None
88. If put command is issued; files are transferred \_\_\_\_\_?
- A. From server to clients
  - B. From clients to server
  - C. Form both side possible
  - D. All
89. Among the following traffic follow which one bidirectional and symmetric?
- A. Peer to Peer
  - B. Server to Sever
  - C. Terminal to Host
  - D. Client to Server
90. An IP workstation can run RIP to learn about routers. RIP should be used in \_\_\_\_\_?
- A. Passive mode
  - B. Active mode
  - C. In both modes
  - D. None

91. Even though sever redundancy is recommended but complete server redundancy is not feasible due to cost issues then what alternative solution you would have?
- A. Duplexing of the file server hard drives
  - B. Mirroring the file servers
  - C. Vitalizing port address
  - D. All
92. From the following alternatives which one is truly related to the term Internet?
- It refers to a collection of two or more computer systems
- A. It is a collection of computer systems and devices that are linked together utilizing LAN, MAN and WAN
  - B. It is the interconnection of a few networks
  - C. It requires fewer hardware devices.
  - D. None
93. WANs (Wide Area Networks) are different from LANs (Local Area Networks) mainly in
- A. WANs have higher bandwidth than LANs
  - B. WANs are more secure than LANs
  - C. WANs covers large size than LANs
  - D. WANs have higher speed than LANs
94. In the OSI reference model of networking which layer relates to message synchronization?
- A. Transport layer
  - B. Application layer
  - C. Presentation layer
  - D. Session layer
95. \_\_\_\_ used to map public IP addresses to private IP addresses in a one-to-one or many-to-one relationships
- A. NAT
  - B. PAT
  - C. TCP
  - D. UD
96. Circuit switching is defined as\_\_\_\_\_?

- A. is used for establishing a dedicated communication path between the sender and the receiver
  - B. Analog telephone is an example for circuit switching
  - C. It has a fixed bandwidth and Data transmission rates
  - D. All
97. High quality voice call uses \_\_\_\_\_?
- A. PSTN
  - B. VoIP
  - C. Circuit switching
  - D. Packet watching
98. The key difference between circuit switching and packet switching is that;
- A. Packet switching is connectionless
  - B. Circuit switching is connection-oriented
  - C. Packet switching is connection-oriented
  - D. A and B
99. Among the following activities which one helps to solve bandwidth problems?
- A. Using data compression mechanism
  - B. Limiting your user and usage
  - C. Buying more bandwidth from the Internet provider
  - D. All
100. In networking Hierarchical design is used to group devices into multiple networks. Therefore, which layer is properly matched with its functions and tasks:
- A. Core Layer – Provides connectivity for network hosts
  - B. Distribution Layer - Interconnects the smaller local networks
  - C. Access Layer - Connects Distribution Layer devices
  - D. All

*Good Luck*